

ADITYA SINGH

Applied AI · LLM Engineering · RAG Architectures · Full-Stack Systems

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PROFILE

Artificial Intelligence & Data Science undergraduate specializing in production LLM architectures, hybrid dense-sparse retrieval systems, and multimodal RAG pipelines. Demonstrated success building resilient enterprise software, high-throughput microservices, and end-to-end full-stack web platforms serving real-world users.

EDUCATION

B.Tech — Artificial Intelligence & Data Science

Vivekananda Institute of Professional Studies (VIPS-TC), GGSIPU · New Delhi

Coursework: Data Structures & Algorithms, Deep Learning, Operating Systems, DBMS, Distributed Systems, Statistics

Expected June 2027

CGPA: 8.8 / 10.0

EXPERIENCE

Software Engineering & AI Intern — Synergy Telecom

May 2024 – Present

- **Full-Stack MERN Architecture:** Engineered production MERN applications with React, Express, and MongoDB, streamlining enterprise inventory management, sales tracking, and operational workflows across internal teams.
- **High-Performance Backend Services:** Designed modular RESTful microservices with JWT authentication, role-based controls, and database indexing, eliminating N+1 query bottlenecks and increasing system throughput under concurrent traffic.
- **Enterprise Hybrid RAG Platform:** Architected an enterprise-grade Retrieval-Augmented Generation system over internal knowledge bases using FastAPI, ChromaDB, and sparse BM25 retrieval with Reciprocal Rank Fusion (RRF).
- **Document Processing & Ingestion Engine:** Built scalable document ingestion pipelines supporting PDFs, Markdown, and HTML with metadata parsing, AST/recursive chunking, and vector embedding generation for grounded QA.

PROJECTS

BIOME AI — Enterprise Knowledge Intelligence System | Python, FastAPI, ChromaDB, BM25, LangChain, Docker

- Architected an end-to-end production RAG engine featuring hybrid retrieval (OpenAI embeddings + BM25) fused via Reciprocal Rank Fusion.
- Integrated cross-encoder reranking, source citation validation, and automated LLM-as-a-Judge benchmarking across 50+ test suites.
- Containerized multi-container microservice infrastructure with Docker Compose and sub-second Streamlit query delivery.

Axiom — Local Multimodal Knowledge Engine | Python, Whisper, ChromaDB, Ollama, PyTorch

[\[GitHub\]](#)

- Developed an offline-first multimodal RAG system indexing video frames, audio transcripts, and technical papers into vector space.
- Implemented local LLM inference via Ollama, completely eliminating cloud API latency and guaranteeing strict zero-data-leak privacy.

StreamPay — Web3 Real-Time Content Monetisation Platform | React.js, Solidity, Monad Testnet, Web3

[\[Live\]](#) | [\[GitHub\]](#)

- Deployed a Web3 content streaming architecture on Monad testnet enabling continuous, real-time creator microtransactions.
- Designed a creator investment bonding pool algorithm allowing audience staking and continuous automated revenue distribution.

Enterprise Operations & Web Platforms | MERN Stack, ImageKit CDN, JWT, Tailwind CSS

[\[Live 1\]](#) | [\[Live 2\]](#)

- Architected and launched production web solutions for clients (Panaceatic Synergy & Glory Education), managing DNS, SSL, and auth pipelines.
- Engineered automated asset tuning via ImageKit CDN and webp transformation, cutting site payload by 80%+ and boosting client engagement by 30%+.

TECHNICAL SKILLS

Core AI & LLM Engineering: RAG Pipelines, Dense + Sparse Search, BM25, Cross-Encoder Reranking, Vector DBs (ChromaDB, FAISS, Qdrant), Fine-Tuning (LoRA), LangChain, LangGraph, LlamaIndex, Transformers, Ollama, OpenAI API

Languages & ML Frameworks: C++ (Primary), Python, JavaScript (ES6+), SQL, Java, PyTorch, Scikit-Learn, Pandas, NumPy

Full-Stack & Distributed Web: React.js, Node.js, Express.js, FastAPI, Redux, Tailwind CSS, REST APIs, WebSockets, Docker, Redis

Databases & Tooling: MongoDB, PostgreSQL, MySQL, Git, Linux (Bash), Postman, Docker Compose, VS Code

ACHIEVEMENTS & CERTIFICATIONS

- **Winner — Cluster Build-a-thon Hackathon:** 1st Place for building an autonomous, end-to-end enterprise AI agent architecture.
- **Competitive Programming:** Solved 300+ DSA problems across LeetCode, Codeforces, and HackerRank.
- **Certifications:** Node.js (Scaler) · Full-Stack Web Development (CodeWithHarry) · Data Science & Python Bootcamp